



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2015
Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

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H2 BIOLOGY

9648/03

Applications Paper and Planning Question

21 September 2015

Paper 3

2 hours

Additional Materials: Answer papers

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough workings.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

At the end of the examination,

1. Fasten all your work securely together.
2. Hand in the following separately:
 - Structured questions 1 – 3
 - Planning question
 - Essay question

The number of marks is given in brackets [] at the end of each question or part question.

ANSWER SCHEME

For examiner's Use	
1	/ 17
2	/ 10
3	/ 13
4 Planning	/ 12
5 Essay	/ 20
Total	/ 72

This paper consists of __ printed pages.

[Turn over]

Answer **all** questions.

QUESTION 1

Golden Rice™ is a genetically modified form of rice that produces relatively large amounts of beta-carotene in the endosperm. Beta-carotene is metabolised in the human body to produce vitamin A.

a) Explain why rice has been genetically modified to produce extra beta-carotene. [2]

- **Ref. to vitamin A deficiency in developing countries**
- **Rice is a staple food / forms a major part of diet in those countries**
- **Increases vitamin A in diet to prevent blindness**

b) The first types of Golden Rice™ produced only a very low mass of beta-carotene per gram of rice. Research continued to try to increase this.

Fig. 1.1 shows the metabolic pathway by which beta-carotene is synthesised in plants, and the enzymes that catalyse each step of the pathway.

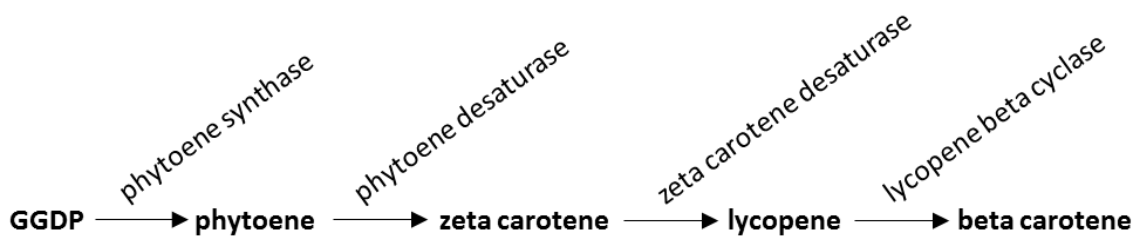


Fig. 1.1

The first types of Golden Rice™ contained a phytoene synthase gene, *psy*, from daffodils and a gene *crtI*, which produced the two desaturase enzymes, from the bacterium *Erwinia uredovora*.

Measurements of the quantities of intermediates in this metabolic pathway in rice endosperm showed that there was always a large amount of GGDP present, and that no phytoene accumulated in the tissues.

Explain how this suggests it was not the enzymes produced by the *crtI* gene that were limiting the production of beta-carotene. [2]

- **GGDP present in large amounts / accumulates / remains high**
 - **Phytoene synthase is limiting because of its low expression / low activity**
- [Reject: non-functional, since the pathway is on-going]**

OR

- **Desaturases are not limiting production because phytoene does not accumulate**
- **Hence, desaturases are functioning normally / converting phytoene to other compounds**

- c) Investigations were carried out to see if *psy* genes taken from species other than daffodils would enable rice endosperm to produce greater quantities of beta-carotene than the first types of Golden Rice™.
- *Psy* genes were isolated from the DNA of maize, tomatoes, peppers and daffodils. The genes were inserted into different plasmids.
 - The promoter *Ubi1* from plants, and *crtI* genes from *E. uredovora*, were also inserted into all of the plasmids.
 - The four types of genetically modified plasmids were then separately inserted into different calluses of rice.
 - The quantity of beta-carotene produced by these rice calluses was measured.

The results are shown in Table 1.1.

Source of <i>psy</i> gene	Total beta-carotene content of rice cells / arbitrary units
maize	14
pepper	4
tomato	6
daffodil	1

Table 1.1

i) Describe the features of calluses. [2]

- An unorganized mass of undifferentiated cells that have no specialized function.
- Totipotent – has the potential to differentiate into specialized plant cells to regenerate the whole plant.
- Rapidly dividing by mitosis to give rise to genetically identical cells.

ii) Outline how calluses of rice were obtained. [2]

- Sterilize the surface of rice explant (e.g. shoot tip) with sodium hypochlorite.
- Culture the explants on a nutrient growth medium with intermediate auxin to cytokinin ratio to induce callus formation by mitosis.

iii) Explain why *Ubi1* promoter was inserted into the plasmids. [2]

- Plasmids are bacterial origin and hence contain prokaryotic promoter
- Eukaryotic promoter *Ubi1* ensures expression of the introduced genes when inside a plant cell.
- **Idea that** eukaryotic transcription factors / RNA polymerase do not recognize prokaryotic promoter

iv) Explain if results in Table 1.1 support the hypothesis that the *psy* gene, not the *crtI* gene, was limiting the production of beta-carotene in genetically modified rice. [2]

- Yes + **data citation** [14au vs 1au / 4au vs 1au / 6au vs 1au]
- The source of the *psy* gene is the only difference
- All rice cells contain the same *crtI* genes. Hence if *crtI* is limiting, there would be no difference in the beta-carotene in each group

d) After the production of the first batch of Golden Rice™ plants, plant tissue culture can continue to be used to generate more Golden Rice™ plants.

i) State one advantage of using plant tissue culture to propagate Golden Rice™ plants. [1]

- Can be produced in large quantity within a short period of time
- Whole plants can be obtained without sexual reproduction, hence all batches of plants are genetically identical to parent plant, retaining desirable traits.
- PTC is an ideal form for export as large numbers can be air freighted easily at low cost to developing / under-developed countries

ii) Suggest why all plant tissue culture methods must be carried out under sterile conditions. [2]

- Prevent contamination with bacteria / fungi which can lead to huge losses.
 - So that plants grow optimally / cells of plant divide more slowly / outgrow by bacteria / competition for nutrients
- [Reject: Viruses (an infection, not contamination), genetically identical (link to diseases)]**

e) An agreement has been made between the commercial company that owns the production rights of golden rice and its developers. This allows the developers to give the rice to government-run breeding centres in rice-dependent countries. This agreement has addressed some concerns that initially arose from growing Golden Rice™.

State these concerns that initially arose from growing Golden Rice™. [2]

- GM seeds could be difficult for farmers in developing countries to obtain.
- GM seeds are too expensive for farmers to buy
- Patenting prevents farmers from re-sowing the GM seed from Golden Rice™ crops
- Erodes farmers' income
- **[consumers]** Prices of such genetically engineered rice crops could be raised
- **AVP**

[Total: 17]

QUESTION 2

Restriction fragment length polymorphism (RFLP) markers identified on the chromosomes are widely used to locate genes along the chromosomes.

In a species of plant, many RFLPs have been mapped within the genome. An RFLP analysis was used to map a herbicide-resistant gene on chromosome 2.

The F_1 offspring from crossing a homozygous herbicide-resistant (R) plant with a homozygous herbicide-sensitive (S) plant were self-fertilized to produce F_2 offspring. The F_2 offspring were analysed with regard to their herbicide sensitivity and two RFLP markers. Each RFLP marker has two variants. The result is shown in Fig. 2.1.

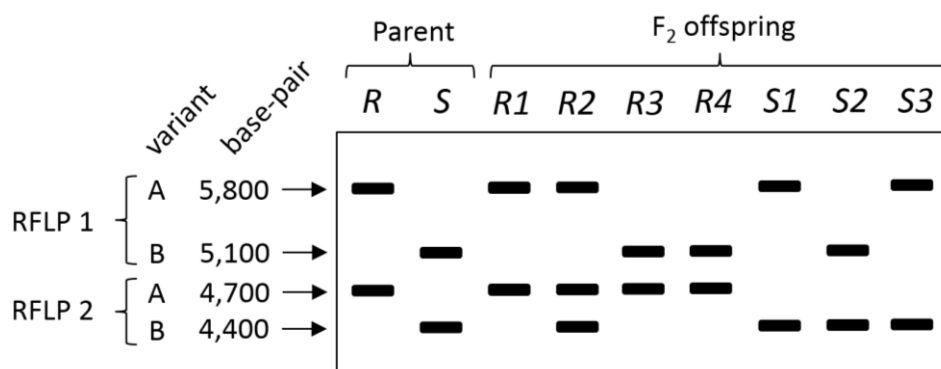


Fig. 2.1

- a) State and explain to which RFLP marker is the gene locus for herbicide-resistance likely to be tightly linked. [2]

- RFLP 2 [Reject: RFLP 2A]
- All resistant plants contains RFLP variant 2A.

OR

All sensitive plants contains RFLP variant 2B.

- b) The distance between two loci is determined by crossing a heterozygote with a homozygote. Fig. 2.2 shows the distance (in terms of map units) between the RFLP loci 1 and 2 on chromosome 2.

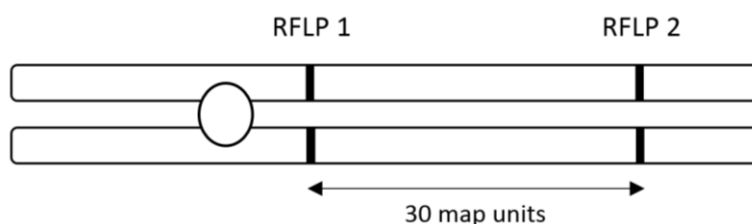


Fig. 2.2

State how the relative distance between the RFLP loci 1 and 2 is determined to be 30 map units. [1]

- The percentage of recombinants is 30%, which translates to a distance of 30 map units / centiMorgans on the chromosome.

In polymerase chain reaction, *Taq* DNA polymerase functions during the elongation step.

c) Outline the role played by other components in a PCR mixture. [3]

- DNA molecule – the two strands are the templates for amplification.
- Deoxyribonucleotides – monomers for polymerization into polynucleotide
- Buffer – provide optimal pH for enzymatic reaction / prevent pH fluctuation
- DNA primers [reject: primer]
 - binds to the 3' regions of the template strands that flank the sequence to be amplified
 - provides a free 3'-OH for *Taq* DNA polymerase to initiate elongation

d) Suggest how data from the Human Genome Project has facilitated the amplification of target DNA sequence using PCR. [1]

- Sequence of nucleotides are known, allowing primers to be designed to bind to sequence that flank the target sequence.

e) Scientists have found a new method of copying DNA that is faster than PCR. The new method, called helicase-dependent amplification (HDA), uses the enzyme helicase to separate the two strands of DNA. This means that DNA can be copied at a constant temperature of 37°C. In all other mechanical aspects, HDA works in exactly the same way as PCR.

i) Explain why HDA will not work with *Taq* DNA polymerase. [2]

- The optimal temperature for *Taq* polymerase is 72°C.
- At a constant temperature of 37°C where HDA is carried out, *Taq* polymerase will be inactive.

ii) Explain why HDA is faster than PCR in amplifying DNA. [1]

- Since HDA operates at a constant temperature, time is not wasted in heating and cooling the mixture.

[Total: 10]

QUESTION 3

Induced pluripotent stem (iPS) cells are adult cells that have been genetically reprogrammed to an embryonic stem cell-like state. Genetic reprogramming is carried out by using modified retroviruses to introduce genes for transcription factors into adult cells from a patient (Fig. 3.1).

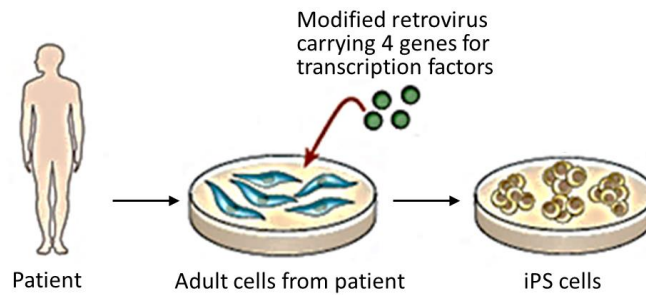


Fig. 3.1

a) Explain the role of the transcription factors in genetic reprogramming of the adult cells. [2]

- The transcription factors can bind to DNA to activate transcription of genes important in maintaining pluripotency.
- E.g. telomerase gene / gene promoting cell division / **AVP**
- Transcription factors also repress the transcription of genes that are important in cell specialisation.

b) Apart from insertion of the four genes for transcription factors, describe two other ways in which the retroviral vector may be modified before they are used for genetic reprogramming of specific adult cells. [2]

- The virulent genes are removed from the viral genome so that they will not cause disease/infect other cells.
- The viral envelope proteins may be modified to bind to specific receptors on the target adult cells.

c) Given that iPS cells are pluripotent like embryonic stem cells, they are promising tools for cell replacement therapy and drug development.

i) Explain two advantages of using iPS cells for cell replacement therapy. [2]

- iPS cells are derived from adult cells of the patient, hence no immune rejection.
- Overcome ethical issues of destroying human embryos to obtain pluripotent stem cells, since iPS cells are derived from adult cells.

ii) Suggest how iPS cells may be used for drug development to treat diseases such as cystic fibrosis. [2]

- iPS cells are induced to differentiate into the lung epithelial cells / **AVP** which carries the mutation.
- Potential **drugs** to treat the disease can be **screened** against this disease tissue, to determine its effectiveness for the specific patient.

Cystic fibrosis is caused by a mutation to the CFTR gene which encodes the Cystic Fibrosis Transmembrane Regulator (CFTR) protein. The CFTR protein regulates the transport of both chloride (Cl^-) and hydrogencarbonate ions (HCO_3^-) across the cell surface membrane. The secretory epithelial cells of the pancreatic duct and airway share the ability to generate HCO_3^- rich fluid.

A study is carried out to determine the effect of 10 different mutant CFTR alleles produced from a point mutation. The transport of HCO_3^- by epithelial cells expressing two copies of the normal CFTR allele was compared with that by epithelial cells expressing two copies of the same mutant CFTR allele. The results are shown in Table 3.1.

In the table, normal digestive functioning of the pancreas associated with a particular allele is indicated by a “+” and absence of normal functioning by a “-”.

CFTR allele		Percentage HCO_3^- transport in comparison with normal allele	Normal digestive functioning of pancreas
Normal		100	+
Mutant	1	6	-
	2	4	-
	3	0	-
	4	3	-
	5	1	-
	6	33	+
	7	41	+
	8	46	+
	9	37	+
	10	44	+

Table 3.1

d) With reference to Table 3.1, describe the differences observed between different mutant CFTR alleles. [2]

- **Mutant alleles 1 to 5** have extremely low percentage HCO_3^- transport compared with normal allele, ranging from 0 to 6% / of **$\leq 6\%$** . Hence the cells containing these alleles **do not have normal digestive functioning** of pancreas.
- **Mutant alleles 6 to 10** have reduced percentage HCO_3^- transport compared with normal allele, ranging from 33 to 46% / of **$\geq 33\%$** . Hence the cells containing these alleles have **normal digestive functioning**.

e) Due to altered HCO_3^- secretion, tissues that express some mutant alleles secrete acidic fluids, while tissues that express normal CFTR alleles secrete alkaline fluids.

i) State the mutant alleles that lead to these acidic secretions. [1]

- Alleles 1 to 5

ii) Briefly explain how the mutated alleles lead to acidic secretions. [2]

- Mutated alleles produce **non-functional CFTR**, which are **unable to transport HCO_3^-** out of the cell.
- Secreted fluid contain **no/minimal HCO_3^-** , which are **alkaline**, hence the fluid is acidic.

[Reject: Silent mutation where same amino acid is added - as none of the mutants have 100% HCO_3^- transport in comparison with normal allele]

[Total: 13]

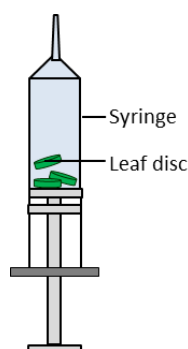
QUESTION 4 – Planning question

You are required to plan, but not carry out an investigation into the effect of carbon dioxide on the rate of photosynthesis.

The spongy mesophyll layer of leaves is normally infused with gases (O_2 and CO_2), and hence discs cut from leaves will float on water. To carry out this experiment effectively, it is important for the leaf discs to sink, which can occur when the gases are forced out from the spongy mesophyll layer using a vacuum. A vacuum can be created by using a finger to close the nozzle of the syringe that is filled with solution. At the same time, draw the plunger to remove any gas that is in the spongy mesophyll layer. Once done, the leaf disc will be able to sink.

In the presence of bicarbonate ions, an alternate source of carbon dioxide, photosynthesis can occur in a sunken leaf disc. As photosynthesis proceeds, oxygen accumulates in the air spaces of the spongy mesophyll, and the leaf disc will once again become buoyant and rise in a column of water. The rate of photosynthesis can therefore be measured. Though some oxygen produced may be used in aerobic respiration, this amount used is negligible and can be ignored.

You are given pieces of fresh leaves, which you need to use along with the set up shown below. Each piece of fresh leaf can only give 10 leaf discs.



(Figure is for illustration purpose only)

Your planning must be based on the assumption that you have been provided with the following apparatus and materials which you must use:

- 8 pieces of fresh spinach leaves
- 0.2 g dm^{-3} Sodium hydrogen carbonate ($NaHCO_3$)
- distilled water
- core borer
- bench lamp which emit little heat
- 5cm^3 syringe
- timer, e.g. stopwatch or stop clock
- plasticine
- meter ruler
- Normal laboratory glassware, e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.

Your plan should:

- have a clear and helpful structure such that the method you used is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with scientific reasoning used to decide the method so that the results areas accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

Planning (Suggested Answers)

Theory and Hypothesis

[Role of CO₂ in photosynthesis] – [1m]

Photosynthesis involves light dependent and light independent reactions. In the light independent reaction, CO₂ is required in carbon fixation where CO₂ combines with ribulose biphosphate to produce 6C intermediate, catalysed by RUBisCO. A higher concentration of CO₂ will increase the rate of carbon fixation and hence the light independent reaction. More ADP and NADP will be regenerated for use in the light dependent reaction.

[Why leaf discs will rise] – [1m]

This in turn increases the rate of light dependent reactions. In the light dependent reactions, photolysis of water occurs to produce electrons to replace those lost in the photosystems during photoactivation. Oxygen gas is produced as by-product during photolysis of water, which will accumulate in the mesophyll cells and increases the buoyancy of the discs, causing them to rise (OWTTE). Time taken for the leaf discs to rise can be used to measure the rate of photosynthesis. The shorter the time taken for the leaf discs to rise, the higher the rate of photosynthesis.

Independent and dependent variables – [1m]

Independent variable: **Concentration of sodium hydrogen carbonate / gdm⁻³**

Dependent variable: **Time taken for the leaf discs to rise / s**

Variables to be kept constant + Rationale – [1m]

These variables are kept constant as they can affect the rate of photosynthesis:

1. Size of the leaf discs – (*Leaf discs prepared using core borer.*)
2. Number of discs - (*State number*)
3. Light intensity – (*Bench Lamp placed specified distance away from experimental set-up.*)
4. Volume of NaHCO₃ – (*Specify volume of NaHCO₃ to be added.*)

[Any 2 variables stated + rationale]

Methods – [2m]

Marking points (3 correct – 2m, 2 correct – 1m)

- 1) Number of leaf discs used per replicate (1–5 only)
 - 2) Dilution table with at least 5 NaHCO_3 concentrations of regular intervals.
 - 3) Creating a vacuum condition to remove air from plant cells, to allow leaf discs to sink + use stopwatch to measure time for leaf to rise. (Accept description quoted from preamble).
 - 4) Acclimatization of leaf discs (1–5 minutes accepted) in different $[\text{NaHCO}_3]$.
 - 5) Keeping 2 stated variables constant.
1. Use a core borer to cut out 80 discs (accept 20 – 80). Keep the leaf discs in the dark.
 2. Prepare 5 different concentration of sodium hydrogen carbonate using the stock solution according to the dilution table shown below:

Concentration of NaHCO_3 / gdm^{-3}	Volume of stock NaHCO_3 / cm^3	Volume of distilled water cm^3	Total volume / cm^3
0.04	4.0	16.0	20
0.08	8.0	12.0	20
0.12	12.0	8.0	20
0.16	16.0	4.0	20
0.20	20.0	0.0	20

3. Remove the plunger from the 5cm^3 syringe and add 5 leaf discs (accept 1-5) into the syringe, taking care not to damage them.
4. Replace the plunger and push it completely into the syringe.
5. Fill the syringe with 5cm^3 of 0.2gdm^{-3} NaHCO_3 . Tap the syringe to suspend the leaf discs in the solution.
6. Place a finger over the nozzle to point the syringe upwards and create a vacuum environment.
7. Draw the plunger to remove any gas present in the leaf cells. Leaf discs will start to sink to the bottom.
8. Repeat steps 6 to 7 until all the leaf discs sink to the bottom.
9. Place the syringe 50cm away from the bench lamp and use a lump of plasticine to hold the syringe in place.
10. Start the stop watch immediately and allow the leaf discs to acclimatise for 3 minutes (accept 1-5min).
11. After 3 minutes, reset the stopwatch and start timing.
12. Note and record the time taken for all the leaf discs to rise to the surface.

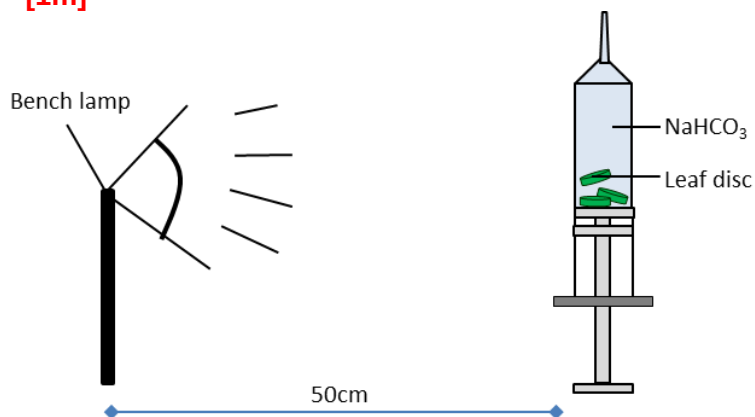
Replicates and Repeats + Rationale – [1m]

13. Repeat steps 3 to 12 to obtain two further readings using the same concentration to minimize error by calculating average/ ensure accuracy.
14. Repeat entire experiment twice, using fresh solutions and leaf samples to ensure reproducibility of the results obtained.

Control – [1m]

- Replace NaHCO_3 solution with equal volume of distilled water.
- Rationale: To show that the rise of leaf disc is due to the oxygen produced when leaf discs undergo photosynthesis.

Labelled diagram – [1m]

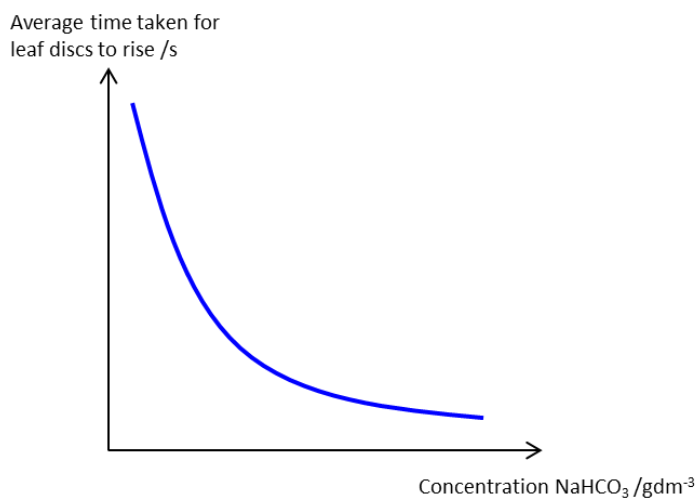


(Not drawn to scale. Mark for stated labels)

Results table – [1m]

Concentration of NaHCO_3 / gdm^{-3}	Time taken for leaf discs to rise / s			
	Reading 1	Reading 2	Reading 3	Average
0.04				
0.08				
0.12				
0.16				
0.20				

Theoretical graph – [1m]



(Graph cannot touch zero)

Risks and precautions – [1m]

Areas of risk	Safety Precautions
NaHCO_3 is an irritant / allergen	Wear gloves when handling it. Wash thoroughly when in contact with skin.
Bench lamp is a source of electrocution.	Ensure hands are dry when handling with bench lamp.
Core borer is sharp and may cause cuts	Handle with care.

Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labeled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a), (b)** etc., as indicated in the question

QUESTION 5

- a) Restriction fragment length polymorphism (RFLP) analysis facilitates the process of DNA fingerprinting. DNA fingerprints can be used for paternity testing and elucidating the murderer in a crime scene.

Outline how DNA fingerprinting is carried out **and** explain its theoretical basis. [12]

[Procedure – max 8]

1. Total genomic DNA extracted from cells / **named cell**
2. Cut by restriction enzyme(s) / **named restriction enzyme**
3. Enzymes chosen to cut restriction sites that flank close to the short tandem repeats
4. Agarose gel electrophoresis to separate restriction fragments
 5. DNA fragments loaded into well at one end of the gel nearer to cathode (and electricity applied)
 6. DNA negatively charged and migrate towards anode
 7. Separation according to molecular size of DNA / Shorter fragments travel further / **reverse argument**
8. Southern blotting – fragments in gel transferred to nitrocellulose membrane
9. NaOH to denature the dsDNA into ssDNA
10. (At least 13) different radioactive single-stranded probes
11. Complementary base pair with the respective STRs
12. Autoradiography / use of X-ray film

[Theoretical basis of DNA fingerprinting – max 4]

13. STRs are non-coding and repeating base sequence
14. Number of repeats differs markedly between individuals
15. Difference in length of DNA fragments between individuals upon restriction digestion is due to the different number of repeats
16. Only identical twins have the same numbers of repeats / genetic fingerprint
17. In diploid organisms, half of the STR is inherited from each parent.

b) Explain why there has been limited success in using gene therapy to treat genetic diseases.

[8]

1. Short-lived nature

- Many vectors **do not** allow for the **integration of the normal allele into the target cell genome** / the **rapidly dividing nature of many cells** (i.e. new cells need to be targeted) prevents the effects of gene therapy from being long-lived and stable.
- As a result, patients have to undergo **multiple rounds of gene therapy**.

2. Immune response

- The **vector** that is introduced into the tissue might be recognised by the immune system as a **foreign** particle (e.g. due to previous infection) and elicit an **immune response**.
- As a result, the effectiveness of the therapy is reduced as the vectors are **destroyed**.

3. Viral vector may regain virulence

- Even though the in the viral vector is modified to be safe/disease-causing viral gene is removed, the viral vector may **regain** or **develop virulence**.
- This causes **disease** in the patient body.

4. Insertional mutagenesis

- For some viral vectors (e.g. retrovirus), the **integration of normal allele into host genome** is **random** and may **disrupt host's gene** or regulatory gene sequences/control elements.
- Disruption of tumour suppressor gene/ conversion of proto-oncogene to oncogene (**one named e.g.**) may cause **cancer**.

5. Multi-gene disorders/disease

- Many common **diseases (one named e.g.** heart disease/high blood pressure/ Alzheimer's disease/diabetes), are caused by the combined effects of **multiple genes**.
- Gene therapy is not effective as it is **impossible to introduce many normal alleles** into target cells at the same time.

6. Large genes

- There may be a **problem finding a suitable vector** if the **normal allele** that needs to be delivered into the target cells is **large** (e.g. viral vectors can only accept genetic material of a certain size).

7. Non-dividing target cells

- Some viral vectors may **not be able to infect non-dividing cells** (eg. retrovirus). Hence gene therapy may not be suitable for diseases involving some cell types.

OR

If the modified target cell (with the normal functional allele) does not divide and multiply, the introduced gene is lost when the cell dies.

8. Diseases caused by presence of a dominant allele

- Gene therapy is ineffective when the **mutated defective allele** is **dominant** and the **normal allele is recessive**.
- The expression of the normal allele will still be **masked** by the dominant defective allele even after being introduced into the target cell genome

9. Problem with controlling the activity of gene expression

- Modified cells **may not synthesize sufficient** functional **proteins** to bring about an improvement.
- Due to (**any valid reason**): e.g. not all cells taking up the normal allele / cells not expressing the proteins at the right time (eg. in thalassemia disease) / normal allele not expressed because they are integrated into a heterochromatic region of the host cell genome.

10. AVP

[1 mark for each well-elaborated point]

[Total: 20]